Spectral Scale Selection in Topological Data Analysis

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Why Select One Scale

Persistent homology is scale-agnostic: the persistence diagram records topological features across all filtration scales simultaneously and thereby avoids committing to any single resolution. Nevertheless, a surprising number of problems in and around TDA reduce to the **choice of one filtration parameter**, and doing so in a theoretically justified way remains a longstanding methodological gap. We list four representative settings in which a topologically justified choice of scale is essential:

Reconstruction and meshing from point clouds

Topology-aware clustering and graph construction

Statistical inference for point processes

Materials and physical systems with intrinsic length scales

Spectral Scale Selection

Let $K_1, \dots, K_N$ be simplicial complexes on a common vertex set X and let $\oplus$ be a commutative, binary operation on them. In what follows, we restrict attention to $\oplus = \cup$. Motivated by **[3]**, we define the f**-based divergence** and the **scale similarity matrix** by

$$\langle K_i, K_j \rangle_f \coloneqq f(K_i) + f(K_j) - f(K_i \oplus K_j) \quad , \quad \Sigma_N \coloneqq \left(\langle K_i, K_j \rangle_f \right)_{i,j=1}^{N} \in \mathbb{R}^{N \times N}.$$

As functional f we consider any combination of **topological or combinatorial features** like Betti numbers or face counts of certain dimension that is -in order for Σ_N to be nonnegative- **subadditive** with respect to the operation $\oplus$. For example, we can choose

$$f(K) = C(K) \cdot E(K),$$

where $C(K)$ is the number of connected components and $E(K)$ the number of edges of K. Since connected components merge and new edges appear in the filtration, the above functional captures the opposing effects and may consequently exhibit **non-monotone** behavior. In the special case where $\{K_i\}_{i=1}^N$ is a filtration, for example $K_i = \mathrm{VR}_{\epsilon_i}(X)$ for some data set X and a grid $0 < \epsilon_1 < \cdots < \epsilon_N$, the f-based divergence reduces to $\langle K_i, K_j \rangle_f = f(K_{\min\{i,j\}})$ and the scale similarity matrix becomes a classical **min/meet-matrix**:

$$\Sigma_N = \begin{pmatrix} f(K_1) & f(K_1) & \cdots & f(K_1) \\ f(K_1) & f(K_2) & \cdots & f(K_2) \\ \vdots & \vdots & \ddots & \vdots \\ f(K_1) & f(K_2) & \cdots & f(K_N) \end{pmatrix}$$

In this easy case, **nonnegativity** of the functional f already implies its subadditivity along the filtration and the scale similarity matrix Σ_N is irreducible iff $f(K_1) > 0$. Consequently, by **Perron-Frobenius**, the principal eigenvector v^* is positive and unique up to scaling and thus induces a ranking of the scales ϵ_i that incorporates their coupling to all other scales through Σ_N. The critical scale is chosen by the rule

$$i_{\mathsf{crit}} \in \arg\max_i v_i^*, \quad \epsilon_{\mathsf{crit}} = \epsilon_{i_{\mathsf{crit}}}$$

Under the above assumptions, it can be shown that the sequences $\{f(K_i)\}_{i=1}^N$ and $\{v_i^*\}_{i=1}^N$ have the same local monotonicity pattern, that is, $\mathrm{sgn}(v_{i+1}^* - v_i^*) = \mathrm{sgn}(f(K_{i+1}) - f(K_i))$ for all $1 \le i \le N-1$ and thus their **local maxima coincide**. For non-monotone functionals, this does not make the Perron vector redundant, since global maxima need not coincide, cf. Figure 1, and the Perron vector can still serve as a **tiebreaker** between separated equal maxima, cf. Figure 2. For strictly monotone functionals f, the Perron vector merely reflects the monotonicity and the critical scale will always lie at an endpoint.

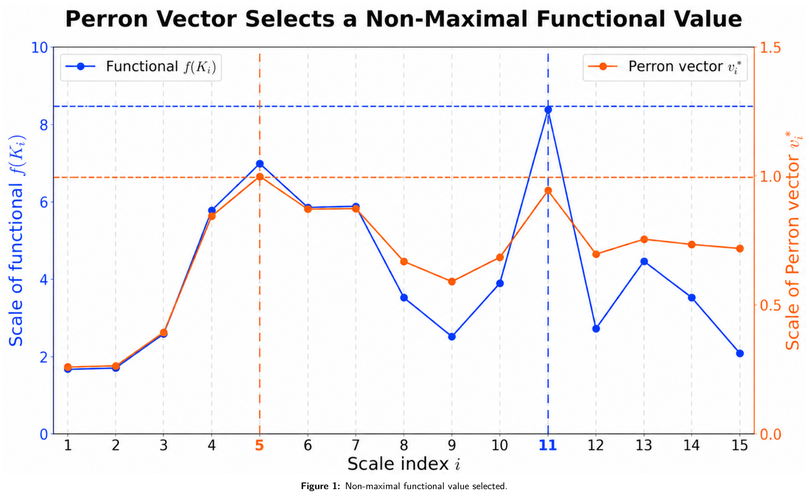


Figure 1: Non-maximal functional value selected.

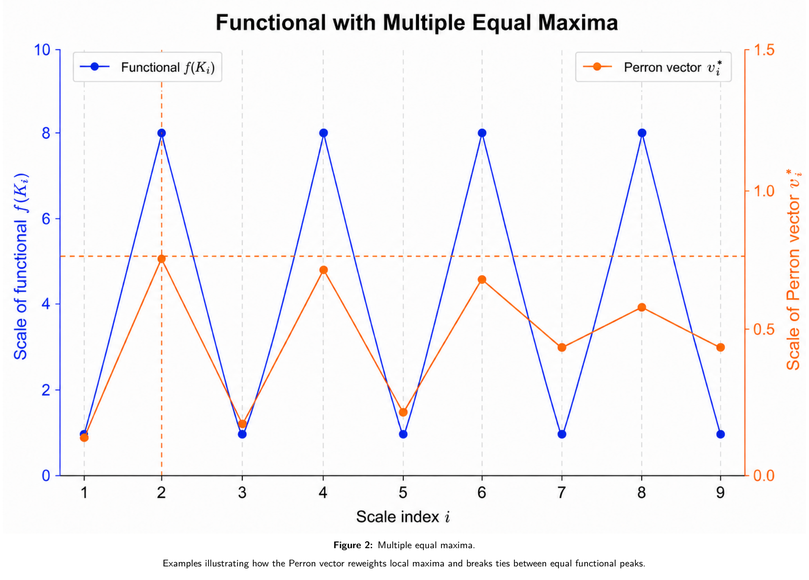


Figure 2: Multiple equal maxima.

Examples illustrating how the Perron vector reweights local maxima and breaks ties between equal functional peaks.

Connection to Markov Chains and Monotone Scale Selection

The spectral scale choice can be motivated by the association of a **finite Markov chain** to the scale similarity matrix Σ_N such that its stationary distribution is proportional to the squared coordinates of the principal eigenvector v^*. Construct the transition matrix P using **Doob's** h**-Transform** and obtain the stationary distribution:

$$P = \frac{1}{\lambda^*} D_{v^*}^{-1} \Sigma_N D_{v^*} \quad , \quad \pi_i^P = \frac{(v_i^*)^2}{\|v^*\|_2^2}$$

where $D_{v^*} = \mathrm{diag}(v_1^*, \dots, v_N^*)$ and $\lambda^* > 0$ the eigenvalue corresponding to the principal eigenvector v^*. The obtained transition matrix P maximizes the **Donsker-Varadhan free-energy functional** over irreducible, row-stochastic matrices Q with stationary distribution π^Q:

$$\mathcal{F}(Q; \Sigma_N) \coloneqq \sum_{i,j=1}^{N} \pi_i^Q q_{ij} \log \frac{(\Sigma_N)_{ij}}{q_{ij}} \quad , \quad P \in \arg\max \left\{ \mathcal{F}(Q; \Sigma_N) : \substack{Q \in \mathbb{R}^{N \times N} \\ \text{row-stochastic and irreducible}} \right\}$$

That is, the transition matrix P is the **variationally optimal stochastic normalization** of the scale similarity matrix Σ_N. The connection to Markov Chains inspires the following approach for strictly increasing functionals, that is, when $0 < f(K_1) < \cdots < f(K_N)$. In such a case, the scale similarity matrix is positive definite and the inverse Σ_N^{-1} is tridiagonal. This allows us to see the **discrete Laplacian** $\mathcal{L} = f(K_1)^{-1}\mathrm{diag}(1, 0, \dots, 0) - \Sigma_N^{-1}$ as the generator of a continuous-time birth-death chain on the state space $\{1, \dots, N\}$. It is given by

$$\mathcal{L} = \begin{pmatrix} -\frac{1}{d_2} & \frac{1}{d_2} & 0 & \cdots & 0 & 0 \\ \frac{1}{d_2} & -\left(\frac{1}{d_2} + \frac{1}{d_3}\right) & \frac{1}{d_3} & \ddots & \vdots & \vdots \\ 0 & \frac{1}{d_3} & -\left(\frac{1}{d_3} + \frac{1}{d_4}\right) & \ddots & 0 & 0 \\ \vdots & \ddots & \ddots & \ddots & \frac{1}{d_{N-1}} & 0 \\ 0 & \cdots & 0 & \frac{1}{d_{N-1}} & -\left(\frac{1}{d_{N-1}} + \frac{1}{d_N}\right) & \frac{1}{d_N} \\ 0 & \cdots & 0 & 0 & \frac{1}{d_N} & -\frac{1}{d_N} \end{pmatrix} \quad , \quad \substack{d_i = f(K_i) - f(K_{i-1}) \\ i = 2, \dots, N}$$

The entries of the transition matrix P^{BD} of the corresponding embedded discrete-time Markov chain are given by $P_{ij}^{\mathsf{BD}} = -(1 - \delta_{ij})\mathcal{L}_{ij}/\mathcal{L}_{ii}$. Its stationary distribution π^{BD} can be used as a scale selection rule for strictly increasing functionals:

$$i_{\mathsf{crit}} \in \arg\min_i \pi_i^{\mathsf{BD}} \quad , \quad \pi_i^{\mathsf{BD}} = \frac{\mathcal{L}_{ii}}{\sum_{j=1}^N \mathcal{L}_{jj}}$$

A visual representation of the monotone scale selection rule is given in Figures 3 and 4. Since $-\mathcal{L}_{ii} = d_i^{-1} + d_{i+1}^{-1}$ for interior scales $2 \le i \le N-1$, minimizing π_i^{BD} favors scales with two large consecutive increments.

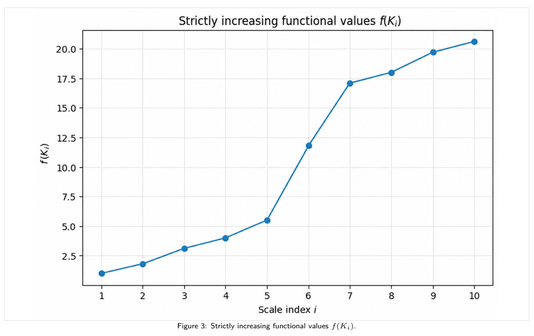


Figure 3: Strictly increasing functional values $f(K_i)$.

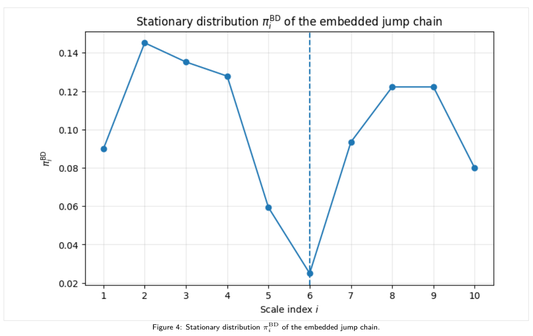


Figure 4: Stationary distribution π_i^{BD} of the embedded jump chain.

The Mapper Algorithm vs. Multiscale α-neighboring (MαN) approach

Let $X = \{x_1, \dots, x_n\} \subseteq \mathbb{R}^D$ be the data at hand, equipped with a metric d. One must first choose a **lens function** $\ell : X \to \mathbb{R}^m$, typically $m \in \{1, 2\}$ and a **finite cover** $\mathcal{U} = \{U_i\}_{i \in I}$ of $\ell(X)$, where $U_i \subseteq \mathbb{R}^m$. The pullback cover is $X_i = \ell^{-1}(U_i)$.

Popular choices for the lens function include:

- density estimate: $\rho(x_i) = \frac{1}{nh^D} \Sigma_{j=1}^n K\left(\frac{d(x_i,x_j)}{h}\right)$ with bandwidth h and K the Gaussian kernel
- centrality: $c(x_i) = \left(n^{-1} \Sigma_{j=1}^n d(x_i, x_j)^p\right)^{1/p}$ for some power $p \ge 1$

Popular choices for the cover include:

- cover in $1D$: interval cover $\{I_j\}_{j=1}^N$ controlled by the number of intervals N and the overlap fraction g
- cover in $2D$: Cartesian product construction $\{U_{ij}\}_{1 \le i \le N_1, 1 \le j \le N_2}$, where $U_{ij} = I_i^{(1)} \times I_j^{(2)}$ with number of intervals N_1, N_2 and overlap fractions g_1, g_2

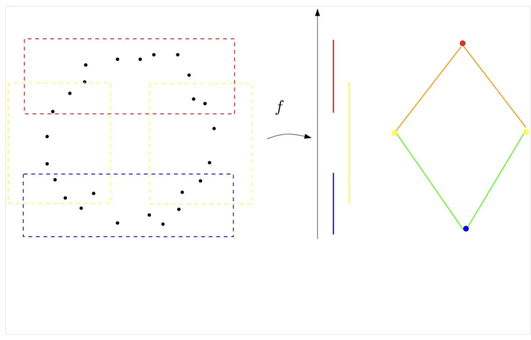


Figure 5: Illustration of Mapper, taken from [1].

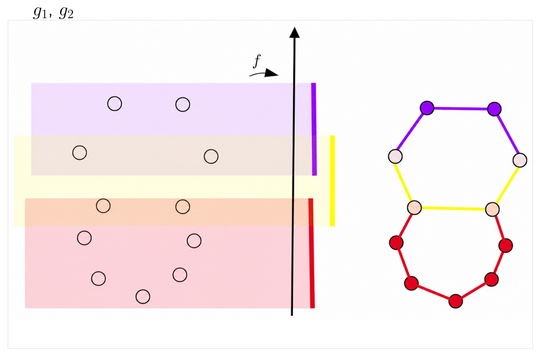


Figure 6: Illustration of the Multiscale α-neighboring graph.

In addition to the lens and cover, the output of Mapper depends on the clustering algorithm used on each X_i. Afterwards, each cluster is seen as a node and an edge between two nodes is formed if the clusters overlap, cf. Figure 5. In some applications, like clinical studies or epidemiological analyses, one wishes to keep the interpretation of each node as a single subject (patient/county), cf. Figure 6. MαN removes clustering from graph construction, but community detection may still be applied afterward. The question becomes, by which rule to connect two nodes.

MαN graph approach: Synthetic dataset experiment

Instead of clustering each X_i, connect two points $x, y \in X$ that belong to at least one common pullback set if and only if

$$d(x, y) \le \max\{2\epsilon_i : i \in I, x, y \in X_i\}$$

The thresholds ϵ_i are chosen by applying the scale selection principle to X_i using the functional $f(K) = C(K) \cdot E(K)$. This allows the graph to **adapt its scale locally** instead of imposing one global threshold. An exemplary scale selection is visualized in Figure 8. As a first stress test, the MαN approach was tested on a **"noisy 8"** dataset.

1. Noisy input X

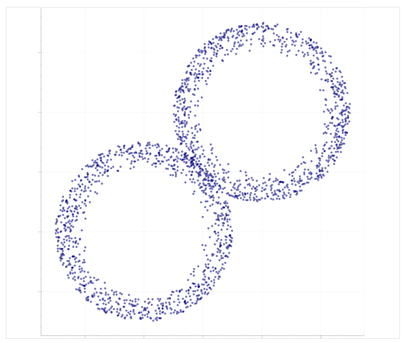


Figure 7: The "Noisy 8" dataset.

2a. Select the local scales ϵ_i

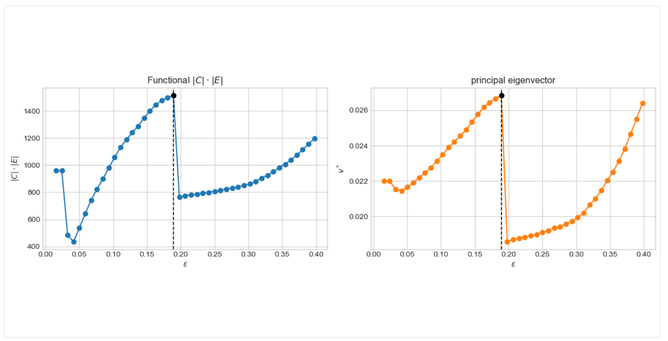


Figure 8: Local scale selection on X_i: the functional profile $j \mapsto f(\mathrm{VR}_{\epsilon_j}(X_i))$ and the corresponding principal eigenvector identify the local threshold ϵ_i.

2b. Grid stability

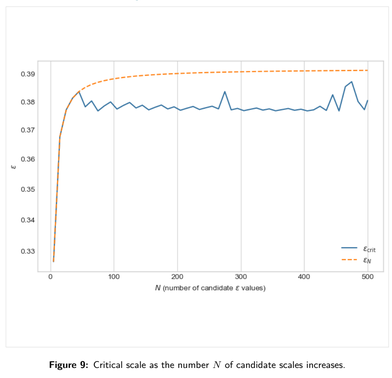


Figure 9: Critical scale as the number N of candidate scales increases.

3a. Random thresholds

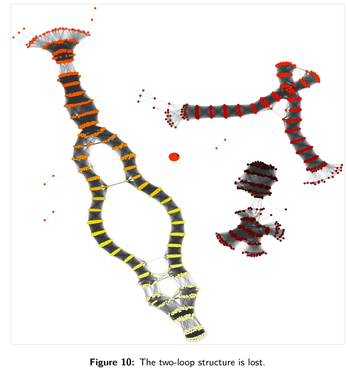


Figure 10: The two-loop structure is lost.

3b. Spectral thresholds



Figure 11: The two-loop structure is recovered.

Case Study: COVID-19 Spread in the USA

Glushakov et al. **[2]** constructed a county-level graph to identify similarities in early COVID-19 trajectories, revealing four major communities associated with demographic and healthcare variables. Here, spectral scale selection determines the local MαN thresholds automatically, using the spectral scale selection with functional $f(K) = C(K) \cdot E(K)$.

County representation

Each node $x_i \in \mathbb{R}^{28}$ represents one of $3{,}142$ US counties and county equivalents.

The first 14 entries encode 13 normalized three-day case aggregates and their log-total; the second 14 analogously encode deaths.

Lens and cover

Lens: $\ell(x) = (\rho(x), c(x))$

Cover: overlapping rectangles $U_{ij} = I_i^{(1)} \times I_j^{(2)}$

Similarity metric

We use the block-wise Euclidean metric on $\mathbb{R}^{14} \times \mathbb{R}^{14}$:

$$d(x_i, x_j) = \|x_i[1{:}14] - x_j[1{:}14]\|_2 + \|x_i[15{:}28] - x_j[15{:}28]\|_2$$

The application of the **Girvan–Newman algorithm** yielded four communities, shown alongside the continuously colored graph in Figures 12 and 13. The coloring in Figure 12 indicates the value of the centrality projection $c(x)$, with darker colors representing higher values, whereas the four colors in Figure 13 indicate the four communities.

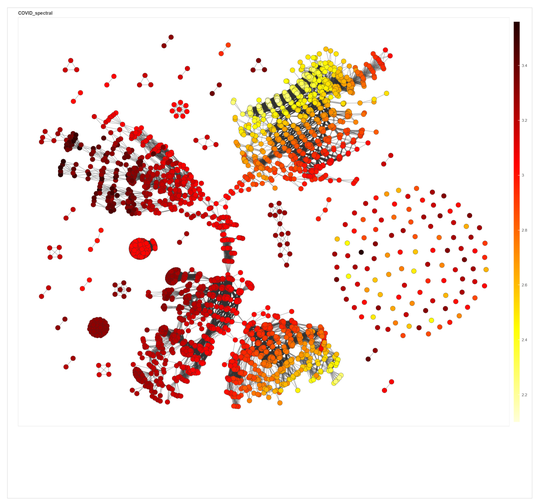


Figure 12: Spectrally selected MαN graph with continuous node coloring.

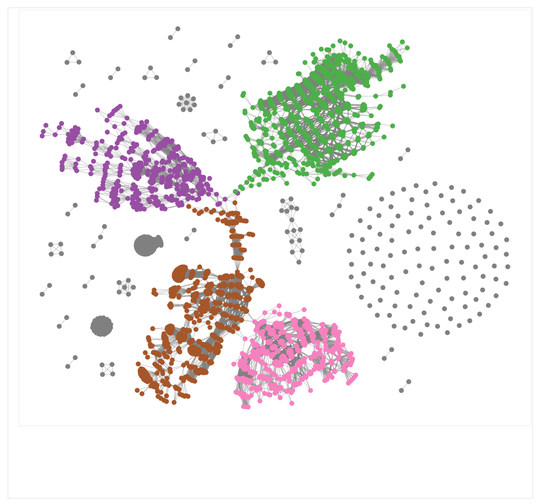


Figure 13: Four communities obtained by Girvan–Newman; remaining nodes are shown in gray.

4
major communities
identified by Girvan–Newman

statistically significant differences
in several county-level predictors not used to build the graph

spectrally selected thresholds
without manual tuning

References

[1] F. Belchí, J. Brodzki, M. Burfitt, and M. Niranjan. *A Numerical Measure of the Instability of Mapper-Type Algorithms.* Journal of Machine Learning Research, 2020.

[2] S. Glushakov, I. Kotenko, and K. Drach. *Understanding the Spread of COVID-19 in the United States Using Topology and Machine Learning for Time Series.* PHUSE EU Connect, 2020.

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